TripNest AI: Test Results

Executed on **2 November 2026** (macOS, Python 3.14.7, pytest 9.2, Playwright 1.58 with Google Chrome, internet available). Everything below was actually run.

Summary

Suite	Command	Passed	Failed	Skipped	Not run
Backend (unit, ML, explore, API)	`cd backend && python -m pytest`	**101**	0	0	0
Browser end to end (real Chrome)	`sh e2e/run.sh`	**24**	0	0	0
Documents	`python docs/build_pdfs.py`	see the build output			
Live deployment checklist	`DEPLOYMENT_GUIDE.md`				**Not run: nothing is deployed**

Backend per file: `test_catalog_crowd.py` 13, `test_core_features.py` 35, `test_crowd_engine.py` 16, `test_explore.py` 28, `test_map_photos.py` 8, `test_schema_upgrade.py` 1. The browser suite checks console errors, page errors and failing API calls in each test.

The nine-step scenario (browser)

Step
Open the app; hero, navigation, search panel, no sign-in
Location: denied permission falls back to a typed city (alias "Bangalore" gives Bengaluru); no permission request on load; granted permission is used at once and the greet
Popular places with photo, crowd chip (green, yellow, red), distance, best time, description, View Details
Date, interest and time of day refresh predictions (Goa on Saturday "High crowd", on Tuesday "Moderate crowd")
Best time to visit: four-slot timeline, calmest slot, hour window, hourly chart; switching destinations updates it
Hidden gems: lesser-known, Low or Moderate, with reasons and "calmer alternative to"
Map: your position, different markers for popular places and gems, filters, popup (photo, level, best time, View Details) fully inside the map
Destination page: prediction, timeline, best time, chart, 7-day tiles, places inside, map, explanation only for the search that produced it
Phone, tablet, laptop and desktop: the flow completes and the page never scrolls sideways

Model evaluation (from `backend/models/metrics.json`, retrained 2 November 2026 on all 271 attractions)

Chronological split; test fortnight 19 October to 1 November 2026 (41,902 hourly rows). Model MAPE **9.38%**, WAPE 8.64%, MAE 54.8 people, R^2 0.986; baseline "same hour last week" MAPE 31.79%; baseline "weekday-hour average" MAPE 27.45%. **Measured on simulated demonstration data only.**

Design inspector and lint (final pages)

App-screen rules at 1440, 768 and 390 px on the home, explore and destination pages. Reviewed and accepted on purpose, because they come from your reference: soft shadows (lint "shadow", 12 places), the Inter typeface ("font-not-dealt", "default-font"), and the cream or blue palette. Remaining notes: icons in tinted circles, large hero headline, uneven card heights beside a sticky side panel, text of a peeking swipe card near the phone edge. No overflow, contrast, headline-line or tiny-text failure remains.

Production-mode checks (run by hand on 2 November 2026, earlier build)

Startup refused with empty or short `SECRET_KEY` and with no `DATABASE_URL`; `alembic upgrade head` created the schema; the server started, `/api/health` was ok and security headers were sent. **Not re-run after the photo-column migration was added**; the migration was generated and applied to an empty database during development.

Defects found and fixed in this redesign

Found by	Defect	Fix
Looking at all 84 photographs	Wrong images: a railway-ministry logo (Darjeeling), a locator map (Daman), a satellite image (Kutch), a night stage (Kohima), a resort signboard (Sohra), a wrong article (Mount Abu)	Better search terms, re-fetched and re-checked
Reading real output	Weak "calmer alternative to" pairings (hill station to a temple 330 km away); gems 700 km away	Specific shared categories only, 200 km limit; gems within 500 km
Looking at the map	A busy lesser-known place drawn as a hidden gem	Only recommended gems get the gem marker; test added
Browser test	Map popup 477 px tall and cut off by the map edge	Compact popup, opened after the map finishes moving
Browser tests at 390 px	Page scrolled sideways (grid sections and timeline columns would not shrink)	Shrinkable grid columns
Phone screenshot	Hero text pushed to the screen edge by the photo credit; header link cut off	Column layout on phones, wrapping header
Phone screenshot	"Why we showed you this" showed reasons from an earlier search	Shown only for the same search and date; test added
Inspector	Hero headline wrapped to four lines on a phone	Sized to fit two
Test authoring	Several of my tests were wrong (CSP-blocked string wait, impossible filter combination, timing races, a no-op assertion)	Corrected; the no-op assertion replaced by a real one

Not tested

Load and performance under many users (single measurements only); PostgreSQL; the Docker image build; the live deployment; browsers other than Chrome; real phones; an accessibility audit with assistive technology; a dependency vulnerability scan; offline use; accuracy on real crowd data.